Swiss Institute of Banking and Finance

University of St.Gallen

# INDEPENDENT STUDIES

## FIRST SESSION

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**3,130 Corporate Finance**

Major in Business Administration (BWL)

# 2. Risk and Return (BMAE, Chapter 7)

## 2.1. True / False Questions

a. The collective of investors is risk averse. This is why all risks will be compensated with a risk premium.

   a. True
   **b. False**

b. Beta is the relation between the unsystematic risk of a single asset and the risk of the market portfolio.

   a. True
   **b. False**

c. Beta measures the relative systematic risk between an asset and the market.

   **a. True**
   b. False

d. A risk premium is the difference between a security's return and the risk-free return.

   **a. True**
   b. False

e. The standard statistical measures of the variability of stock returns are beta and covariance.

   a. True
   **b. False**

f. Diversification reduces the risk of a portfolio when prices of different securities do not move exactly together.

   **a. True**
   b. False

g. The portfolio risk that cannot be eliminated by diversification is called specific or idiosyncratic risk.

   a. True
   **b. False**

h. The average beta of all stocks in the market is zero.

   a. True
   **b. False**

i. The beta of an equally-weighted portfolio including all assets is equal to 1.
   a. True
   **b. False**

j. A portfolio with a beta of one offers an expected return equal to the market risk premium.

   a. True
   **b. False**

k. Stocks with high standard deviations will necessarily also have high betas.

   a. True
   **b. False**

l. A stock having a covariance with the market that is higher than the variance of the market will always have a beta above 1.0.

   **a. True**
   b. False

m. The variability of a well-diversified portfolio mostly reflects the contributions to risk from the standard deviations of the stocks within that portfolio.

   a. True
   **b. False**

n. Two stocks can never have a positive correlation and a negative covariance.

   **a. True**
   b. False

o. The covariance between the returns on two stocks equals the correlation coefficient multiplied by the standard deviations of the two stocks.

   **a. True**
   b. False

p. According to Fisher (1907), the relation between real and nominal interest rates and inflation is $(1 + r_{nominal}) = (1 + r_{real})*(1 + \text{inflation rate})$.

   **a. True**
   b. False

q. The average inflation rate should be added to the average annual nominal return of Treasury Bills in order to find the average annual real return of Treasury Bills.

   a. True
   **b. False**

## 2.2. Market risk premium

The average annual rate of return for common stocks is 11.7% with a standard deviation of 10%. The average compound annual return is 10.5% with a standard deviation of 9.5%. The risk-free interest rate is 4.0%. What is the most likely average market risk premium under the assumption that one can apply historical returns to forecast future expected returns?

a. 4.0%
b. 6.5%
**c. 7.7%**
d. 14.5%
e. 15.7%

$$RP_{Market} = 0.117 - 0.04 = 0.077$$

## 2.3. Stock prices

An investor buys one stock of firm A for CHF 600. One year later, the stock price has increased by 10%. In the following years, the investor can achieve returns of 20% (year 2), -20% (year 3), and 10% (year 4). What is the most likely stock price after year 4?

a. CHF 174.24
b. CHF 633.60
**c. CHF 696.96**
d. CHF 720.00
e. CHF 732.19

$$P_4 = 600 \cdot (1 + 0.1) \cdot (1 + 0.2) \cdot (1 - 0.2) \cdot (1 + 0.1) = 696.96$$

## 2.4. Expected return

Stock A has an expected return of 10% p.a. and stock B has an expected return of 20% p.a. If 40% of a portfolio's funds are invested in stock A, and the rest in stock B, what is the most likely expected return on the portfolio of stock A and stock B?

a. 10% p.a.
b. 14% p.a.
**c. 16% p.a.**
d. 18% p.a.
e. 20% p.a.

$$E(r) = 0.1 \cdot 0.4 + 0.2 \cdot 0.6 = 0.16$$

## 2.5. Standard deviation of stock returns

Mega Corporation showed the following returns in the past three years: 8%, 12%, and 10% which is only a sample of a longer time series. What is the most likely estimated standard deviation of the returns?

**a. 2.0%**
b. 3.4%

c. 8.0%
d. 10.0%
e. 11.1%

$$\mu_r = \frac{0.08 + 0.12 + 0.1}{3} = 0.1$$

$$Var = \frac{1}{3-1} \cdot \sum_{t=1}^{3} (r_t - 0.1)^2 = \frac{1}{2} \cdot ((0.08 - 0.1)^2 + (0.12 - 0.1)^2 + (0.1 - 0.1)^2) = 0.0004$$

$$\sigma = \sqrt{Var} = \sqrt{0.0004} = 0.02$$

## 2.6. Comovement

A statistical measure of the degree to which securities' returns move together is called:

a. Variance
**b. Covariance**
c. Standard deviation
d. Geometric average
e. None of the answers a. to d. are correct.

## 2.7. Market risk

Market risk is also called: I) systematic risk; II) undiversifiable risk; III) firm-specific risk.

a. I only
b. II only
c. III only
**d. I and II only**
e. II and III only

## 2.8. Number of stocks in a portfolio

As the number of uncorrelated stocks in a portfolio increases:

**a. Firm-specific risk decreases and approaches zero**
b. Market risk decreases
c. Firm-specific risk decreases and becomes equal to market risk
d. Total risk approaches zero
e. None of the answers a. to d. are correct.

## 2.9. Covariance

Stock A and Stock B showed the following returns in the past three years: -10%, 12%, 28% and -5%, 14%, 24%, respectively. The reported returns are both only samples from longer time series. What is the most likely estimated covariance between the two securities?

a. -2.80
b. -1.25
c. 1.25
d. 2.20
**e. 2.80**

$$E[\tilde{A}] = \frac{-0.10 + 0.12 + 0.28}{3} = 0.1$$

$$E[\tilde{B}] = \frac{-0.05 + 0.14 + 0.24}{3} = 0.11$$

$$Cov[A_t, B_t] = \frac{1}{3-1} \cdot \sum_{t=1}^{3} \left( \left(A_t - E[\tilde{A}]\right) \cdot \left(B_t - E[\tilde{B}]\right) \right)$$

$$Cov[A_t, B_t] = \frac{1}{2} \cdot \left( (-0.10 - 0.1) \cdot (-0.05 - 0.11) + (0.12 - 0.1) \cdot (0.14 - 0.11) + (0.28 - 0.1) \cdot (0.24 - 0.11) \right)$$
$$= 0.028$$

**2.9.1.**
Assume that the correlation coefficient between the returns on stock C and stock D is +1.0, the standard deviation of returns for stock C is 20%, and that for stock D is 30%. What is the most likely covariance between stock C and stock D?

a. -0.60
b. -0.06
**c. 0.06**
d. 0.36
e. 0.60

$$Cov = \sigma_C \cdot \sigma_D \cdot \rho_{C,D} = 0.20 \cdot 0.3 \cdot 1 = 0.06$$

**2.9.2.**
What range of values can correlation coefficients take?

a. zero to + 1
**b. -1 to + 1**
c. -infinity to + infinity
d. zero to + infinity
e. -infinity to +1

**2.9.3.**
Assume that the covariance between stock A and stock B is 0.01, the standard deviation of stock A is 10% and that of stock B is 20%. What is the most likely correlation coefficient between both securities?

a. -0.5
b. -0.25
c. 0
d. 0.25
**e. 0.5**

$$Cov = \sigma_C \cdot \sigma_D \cdot \rho_{C,D}$$

$$0.01 = 0.1 \cdot 0.2 \cdot \rho_{A,B} \rightarrow \rho_{A,B} = 0.5$$

**2.9.4.**
For a two-stock portfolio and no short selling, the maximum reduction in risk occurs when the correlation coefficient between the two stocks equals:

**a. -1.0**
b. -0.5
c. 0
d. 0.5
e. 1.0

## 2.10. Beta

**2.10.1.**
The Beta of the market portfolio is:

a. -1.0
b. -0.5
c. 0.0
d. 0.5
**e. 1.0**

**2.10.2.**
Beta is a measure of:

a. Total risk
b. Idiosyncratic risk
**c. Systematic risk**
d. Liquidity risk
e. None of the answers a. to d. are correct.

**2.10.3.**
Which of the following portfolios had the highest historical beta according to the study by Dimson, Marsh, and Staunton (2002)?

a. Portfolio of U.S. Treasury bills
b. Portfolio of U.S. government bonds
c. Portfolio containing 50% U.S. Treasury bills and 50% U.S. government bonds
**d. Portfolio of U.S. common stocks**
e. None of the answers a. to d. are correct.

**2.10.4.**
Historical returns for stock A were -8%, +10%, and +22%. Returns for the market portfolio (m) were +6%, +18%, and 24% during the same period. Both time series are samples of longer time series. What is the most likely estimated beta for stock?

a. 0.50
b. 0.61
c. 1.00
**d. 1.64**
e. 1.78

$$E[\tilde{A}] = \frac{-0.08 + 0.1 + 0.22}{3} = 0.08$$

$$E[\tilde{m}] = \frac{0.06 + 0.18 + 0.24}{3} = 0.16$$

$$Var(m) = \frac{1}{3-1} \cdot ((0.06 - 0.16)^2 + (0.18 - 0.16)^2 + (0.24 - 0.16)^2) = 0.0084$$

$$Cov[A_t, m_t] = \frac{1}{2} \cdot \big((-0.08 - 0.08) \cdot (0.06 - 0.16) + (0.1 - 0.08) \cdot (0.18 - 0.16) + (0.22 - 0.08) \cdot (0.24 - 0.16)\big)$$
=0.0138

$$\beta_A = \frac{0.0138}{0.0084} = 1.64$$

**2.10.5.**
The covariance between stock C and the S&P 500 is 0.05. The standard deviation of the stock market is 20% and of the stock is 35%. What is the most likely beta of C?

a. 0.05
b. 0.14
c. 0.41
**d. 1.25**
e. 1.36

$$\beta_C = \frac{0.05}{0.2^2} = 1.25$$

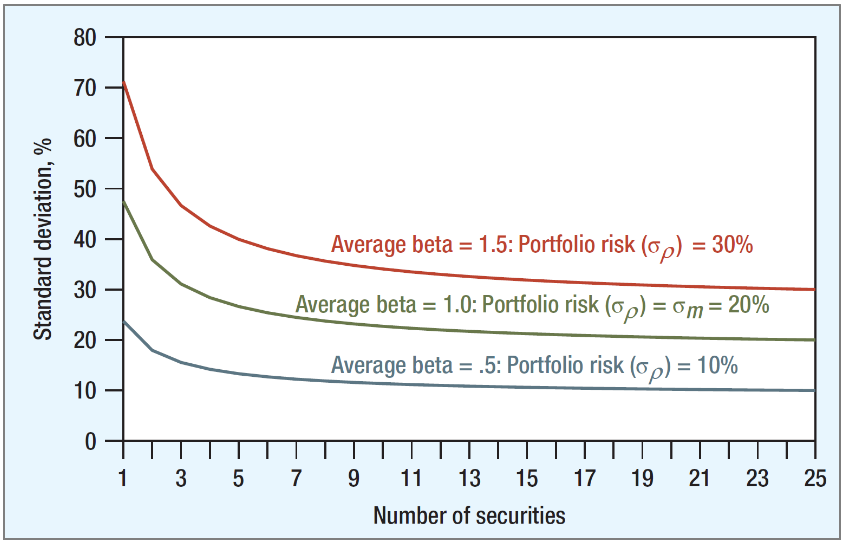
**2.10.6.**
What is the beta of a security that has a zero-correlation with the U.S. stock market and whose expected return is double that of the U.S. stock market? The standard deviation of the stock market is .18?

**a. 0.00**
b. 1.00
c. 1.25
d. 2.00
e. 2.50

$$\beta = \frac{0.00}{0.18^2} = 0$$

## 2.11. Portfolio Diversification

Which statement is false about the graph below displaying three well-diversified portfolios of randomly selected stocks?




a. The upper red line shows that a well-diversified portfolio with β = 1.5 has a standard deviation of about 30% i.e.,1.5 times that of the market.
**b. The standard deviation of each of the three portfolios tends to decrease as this portfolio contains less and less securities.**
c. The green line shows that a well-diversified portfolio of randomly selected stocks ends up with β = 1 and a standard deviation equal to the market's, i.e., in this case 20%.
d. The lower blue line shows that a well-diversified portfolio with β = .5 has a standard deviation of about 10%, i.e., half that of the market.
e. Each of these three portfolios is composed of securities only.

# 3. Portfolio Theory and CAPM (BMAE, Chapter 8)

## 3.1. True / False Questions

a. When measured over monthly intervals, the past rates of return on stocks generally conform fairly closely to a normal distribution.

   **a. True**
   b. False

b. Because the normal distribution is skewed it can be described by only two moments (mean and variance).

   a. True
   **b. False**

**c.** The return of an asset located below the Security Market Line is expected to increase as such an asset outperforms the market.

   a. True
   **b. False**

d. A lower portfolio risk than the average risk of two separate assets indicates a correlation coefficient lower than 1.

   **a. True**
   b. False

e. Diversification cannot reduce portfolio risk if correlations across stock returns equal zero.

   a. True
   **b. False**

f. Two assets with a perfectly negative correlation and positive returns can never make up a portfolio that has a variance of zero but a positive return.

   a. True
   **b. False**

g. Efficient portfolios are the combinations of risky assets which offer the highest expected return for any level of risk.

   **a. True**
   b. False

h. Borrowing money at the risk free rate to over-invest in the market portfolio allows to increase expected return but decrease risk exposure.

   a. True
   **b. False**

i. The CAPM makes a statement about expected returns.

a. **True**
b. False

j. Assets with a beta of zero are free of systematic risk.

a. **True**
b. False

k. The CAPM states that a stock with a beta of zero will have an expected return of zero.

a. True
b. **False**

l. The CAPM assumes that information is not costly.

a. **True**
b. False

m. In a CAPM world, the factor loading on market risk is the only relevant risk sensitivity.

a. **True**
b. False

n. The CAPM can only be used to predict future expected returns of a firm if the firm is well-diversified.

a. True
b. **False**

o. The beta of a well-diversified portfolio is equal to the value weighted average beta of the securities included in the portfolio.

a. **True**
b. False

p. Investors demand a higher expected return for stocks that respond very sensitively to changes in the capital market.

a. **True**
b. False

q. CAPM is a model postulating that an asset's expected return is proportional to a measure of the asset's systematic risk.

a. **True**
b. False

r. According to the CAPM, the risk premium on a security is directly proportional to its covariance with the market portfolio.

a. **True**
b. False

s. If the CAPM holds, stocks which are below the security market line should be sold as they are considered to be overvalued.

   a. **True**
   b. False

t. The Sharpe Ratio defines the excess return per unit of systematic risk.

   a. True
   b. **False**

u. The three risk factors of Fama and French are market risk, momentum, and firm size.

   a. True
   b. **False**

### 3.2. Normal distribution

A normal distribution can be fully characterized by: I) skewness; II) variance; III) mean; IV) kurtosis.

a. I and II
b. II and III and IV
c. I and IV
**d. II and III**
e. III and IV

### 3.3. Beta

The market consists of 2 companies with the following relative size distribution: 30% Rolling Ltd., 70% Stones Inc.

| | Rolling Ltd. | Stones Inc. |
|---|---|---|
| Standard Deviation | 20% | 40% |
| Covariance (Rolling, Stones) | 0.08 | 0.08 |

The beta of Rolling Ltd. equals most likely

**a. 0.59**
b. 0.69
c. 1.00
d. 1.45
e. 1.69

$$\beta_{Rolling} = \frac{Cov(Rolling, market)}{Var(market)}$$

$$Var(market) = (0.3 \cdot 0.2)^2 + (0.7 \cdot 0.4)^2 + 2 \cdot 0.08 \cdot 0.3 \cdot 0.7 = 0.1156$$

$$Cov(Rolling, market) = Cov(Rolling, 0.3 \cdot Rolling + 0.7 \cdot Stones)$$
$$Cov(Rolling, market) = 0.3 \cdot Cov(Rolling, Rolling) + 0.7 \cdot Cov(Rolling, Stones)$$
$$Cov(Rolling, market) = 0.3 \cdot Var(Rolling) + 0.7 \cdot Cov(Rolling, Stones)$$
$$Cov(Rolling, market) = 0.3 \cdot 0.2^2 + 0.7 \cdot 0.08 = 0.068$$

$$\beta_{Rolling} = \frac{0.068}{0.1156} = 0.588$$

## 3.4. Firm Beta

The following information about X-Corporation is given:

| | |
|---|---|
| Expected risk premium of X-Corporation | 12% |
| Standard deviation of market portfolio | 21% |
| Sharpe ratio of market portfolio | 38% |

The most likely beta of the X-Corporation is …

a. 1.0
b. 1.4
**c. 1.5**
d. 1.6
e. None of the answers a. to d. are correct.

$$Sharpe\ ratio_{market} = \frac{E(r_m) - r_f}{\sigma_m} = \frac{risk\ premium_m}{\sigma_m}$$

$$0.38 = \frac{risk\ premium_m}{0.21} \rightarrow risk\ premium_m = 0.0798$$

$$E(r_X) - r_f = \beta \cdot \left(E(r_m) - r_f\right)$$

$$0.12 = \beta \cdot 0.0798 \rightarrow \beta = 1.5$$

## 3.5. Expected Return and Risk

Which statement is correct?

**a. The relation between expected return and risk, measured by beta, is linear.**
b. The unsystematic risk in the capital market is measured by beta.
c. Beta of the market portfolio is on average zero.
d. The Beta of a stock is distributed log-normally.
e. None of the answers a. to d. are correct.

## 3.6. Efficient frontier

You find the following correlation matrix and expected return vector for a market with three assets (asset 1, asset 2, asset 3).

$$\begin{pmatrix} \sigma_1^2 & \rho_{12} & \rho_{13} \\ \rho_{21} & \sigma_2^2 & \rho_{23} \\ \rho_{31} & \rho_{32} & \sigma_3^2 \end{pmatrix} = \begin{pmatrix} 0.3 & 1 & 1 \\ 1 & 0.1 & 0.2 \\ 1 & 0.2 & 0.2 \end{pmatrix} \quad , \quad \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix} = \begin{pmatrix} 0.15 \\ 0.1 \\ 0.2 \end{pmatrix}$$

Which assets will be included in any efficient portfolio?

a. All three assets.
b. Only asset 1 and/or 2.
**c. Only asset 2 and/or 3.**

d. Only asset 1 and/or 3.
e. None of the above.

### 3.7. Investments with borrowing and lending

Suppose that the market portfolio has an expected return of 11% and a standard deviation of 13%. Treasury bills offer an interest rate of 5% and are risk-free. In order to have less risk exposure than the market portfolio, you invest 70% of your money in the market portfolio and lend the remainder at the risk-free rate. What are the mostly likely expected return and standard deviation on this investment?

a. The expected return is 8.2% and the standard deviation is 7.1%.
b. The expected return is 8.6% and the standard deviation is 9.1%.
**c. The expected return is 9.2% and the standard deviation is 9.1%.**
d. The expected return is 9.5% and the standard deviation is 6.5%.
e. The expected return is 10.3% and the standard deviation is 13.5%.

$$r = 0.7 \cdot 0.11 + 0.3 \cdot 0.05 = 0.092$$

$$\sigma = \sqrt{(0.7 \cdot 0.13)^2 + (0.3 \cdot 0.00)^2 + 2 \cdot 0.7 \cdot 0.3 \cdot 0.13 \cdot 0 \cdot 0} = 0.091$$

Since the correlation between the risk-free rate and any portfolio is 0, the formula for the standard deviation could be simplified to (as we did on slide 15, lecture 2):

$$\sigma = 0.7 \cdot 0.13 + 0.3 \cdot 0.00 = 0.091$$

### 3.8. CAPM

The risk-free interest rate is 4% and the market risk premium is 6%. A large bank determines on the basis of the CAPM a return of 15% for the stock X. What was the most likely beta which the bank used in order to calculate the return of stock X?

a. The beta of stock X is 1.55.
b. The beta of stock X is 1.74.
**c. The beta of stock X is 1.83.**
d. The beta of stock X is 2.00.
e. None of the answers a. to d. are correct.

$$E(r_X) = r_f + \beta \cdot \left(E(r_m) - r_f\right)$$

$$0.15 = 0.04 + \beta \cdot 0.06 \rightarrow \beta = 1.83$$

## 3.9. Sharpe Ratio

The risk-free rate is 1%, the market risk-premium is 6%, the variance of the market is 0.0625. Sort the following assets by their Sharpe Ratio.

| Asset | Beta | Variance |
|---|---|---|
| A | 2.0 | 0.16 |
| B | 1.5 | 0.25 |
| C | 1.0 | 0.04 |

a. Sharpe Ratio A > Sharpe Ratio B > Sharpe Ratio C
b. Sharpe Ratio B = Sharpe Ratio A > Sharpe Ratio C
c. Sharpe Ratio B > Sharpe Ratio A = Sharpe Ratio C
**d. Sharpe Ratio A = Sharpe Ratio C > Sharpe Ratio B**
e. None of the answers a. to d. are correct.

$$Sharpe\ ratio_A = \frac{E(r_A) - r_f}{\sigma_A} = \frac{0.01 + 2 \cdot 0.06 - 0.01}{\sqrt{0.16}} = 0.3$$

$$Sharpe\ ratio_B = \frac{E(r_B) - r_f}{\sigma_B} = \frac{0.01 + 1.5 \cdot 0.06 - 0.01}{\sqrt{0.25}} = 0.18$$

$$Sharpe\ ratio_C = \frac{E(r_C) - r_f}{\sigma_C} = \frac{0.01 + 1 \cdot 0.06 - 0.01}{\sqrt{0.04}} = 0.3$$

## 3.10. Arbitrage Pricing Theory

On the market, you find the following assets: A, C, DC. From your research, you know that the risk-free rate equals 1%, the expected market return equals 6%, SMB factor equals 2%, and HML factor equals 4%. Sort A, C, DC by their expected performance.

| Asset | Beta MKT | Beta SMB | Beta HML |
|---|---|---|---|
| A | 2.0 | -0.16 | 1.0 |
| C | 1.5 | 1.0 | 0.8 |
| DC | 1.0 | 1.4 | 0.9 |

a. **$E[R_A] > E[R_C] > E[R_{DC}]$**
b. $E[R_C] > E[R_{DC}] > E[R_A]$
c. $E[R_{DC}] > E[R_A] > E[R_C]$
d. $E[R_A] > E[R_C] = E[R_{DC}]$
e. None of the answers a. to d. are correct.

$$E(r_A) = 0.01 + 2 \cdot (0.06 - 0.01) - 0.16 \cdot 0.02 + 1 \cdot 0.04 = 0.1468$$

$$E(r_C) = 0.01 + 1.5 \cdot (0.06 - 0.01) + 1 \cdot 0.02 + 0.8 \cdot 0.04 = 0.137$$

$$E(r_{DC}) = 0.01 + 1 \cdot (0.06 - 0.01) + 1.4 \cdot 0.02 + 0.9 \cdot 0.04 = 0.124$$

## 3.11. Arbitrage Pricing Theory and investment strategy

Investments X, Y, and Z have the following sensitivities to two macroeconomic factors.

| Investment | Beta 1 | Beta 2 |
|---|---|---|
| X | 1.75 | 0.25 |
| Y | -1.00 | 2.00 |
| Z | 2.00 | 1.00 |

We assume that the expected risk premium is 4% on factor 1 and 8% on factor 2. Treasury bills obviously offer zero risk premium. Suppose that you buy USD 200 of X and USD 50 of Y and sell USD 150 of Z. What is the expected risk premium of your portfolio based on an arbitrage pricing theory model which uses these two macroeconomic factors?

a. – 10%
b. – 5%
c. **0%**
d. 5%
e. 10%

$E(r_X) = 1.75 \cdot 0.04 + 0.25 \cdot 0.08 = 0.09$
$E(r_Y) = -1.00 \cdot 0.04 + 2.00 \cdot 0.08 = 0.12$
$E(r_Z) = 2.00 \cdot 0.04 + 1.00 \cdot 0.08 = 0.16$

$200 \cdot 0.09 + 50 \cdot 0.12 - 150 \cdot 0.16 = 0$

### 3.12. Change of Beta

The stock of Y-Corporation has a beta of 1.3. Which statement is correct?
If the market return increases by 200 basis points, …

a. the return of stock Y remains constant.
b. the return of stock Y increases also by 200 basis points.
c. the return of stock Y decreases by 260 basis points.
d. the return of stock Y decreases by 200 basis points.
**e. None of the answers a. to d. are correct.**

$$\Delta r_x = \beta \cdot \Delta r_m = 1.3 \cdot 200\ bps = +260\ bps$$

Remark: 100 basis points equal 1%.

### 3.13. Time stability of Beta

Empirical studies show that the correlations between individual stocks and the market portfolio are not stable but change over time. Assume that the correlation between stock X and the market portfolio doubles in a certain time-frame.

What is the impact (ceteris paribus) on Beta?

a. The Beta doesn't change.
**b. The Beta duplicates.**
c. The Beta divides by two.
d. A statement about the impact on Beta is not possible as further information is necessary.
e. None of the answers a. to d. are correct.

$$\beta = \frac{\sigma_X \cdot \sigma_m \cdot \rho_{X,m}}{\sigma_m{}^2} \rightarrow 2 \cdot \rho_{X,m} \rightarrow \beta = \frac{\sigma_X \cdot \sigma_m \cdot (2 \cdot \rho_{X,m})}{\sigma_m{}^2} \rightarrow 2 \cdot \beta$$

### 3.14. Nominal and real interest rates

Mr. X invests USD 1,000 at a 10% nominal rate for one year. If the inflation rate is 4% p.a., what is the purchasing power of the investment at the end of one year?

a. USD 945.45
b. USD 1,000.00
c. USD 1,040.00
**d. USD 1,057.69**
e. USD 1,100.00

$$(1 + r_{nominal}) = (1 + r_{real}) \cdot (1 + inf.)$$

$$r_{real} = \frac{(1 + r_{nominal})}{(1 + inf.)} - 1 \approx 0.0577 = 5.77\%$$

$$Real\ value = 1{,}000 \cdot (1 + r_{real}) = 1{,}000 \cdot (1 + 0.0577) \approx 1{,}057.69$$

## 3.15. Inflation, nominal and real returns

Suppose that the nominal average rate of return p.a. is 3.8% on Treasury Bills, 5.3% on Government Bonds and 11.5% on Common stocks. The average inflation rate is 3% p.a. What is the most likely real average rate of return p.a. for each of these 3 assets? What is the average risk premium p.a., i.e. extra return versus Treasury Bills, for each of these 3 assets?

a. The real average rate of return is: 3.8% for Treasury Bills, 5.3% for Government bonds and 11.5% on Common stocks; The average risk premium is: 0% for Treasury Bills, 1.5% for Government bonds and 7.7% on Common stocks.
b. The real average rate of return is: 6.8% for Treasury Bills, 8.3% for Government bonds and 14.5% on Common stocks; The average risk premium is: 0% for Treasury Bills, 1.5% for Government bonds and 11.5% on Common stocks.
c. The real average rate of return is: 1.8% for Treasury Bills, 3.3% for Government bonds and 9.5% on Common stocks; The average risk premium is: - 1.8% for Treasury Bills, 1.3% for Government bonds and 7.5% on Common stocks.
d. The real average rate of return is: 5.8% for Treasury Bills, 7.3% for Government bonds and 13.5% on Common stocks; The average risk premium is: 7.8% for Treasury Bills, 9.3% for Government bonds and 15.5% on Common stocks.
**e. The real average rate of return is: 0.8% for Treasury Bills, 2.2% for Government bonds and 8.3% on Common stocks; The average risk premium is: 0% for Treasury Bills, 1.5% for Government bonds and 7.7% on Common stocks.**

| | Nominal average annual rate of return | Real average annual rate of return | Average risk premium (extra return vs. Treasury bills) |
|---|---|---|---|
| Treasury bills | 3.8% | 0.8% | 0% (1) |
| Government bonds | 5.3% | 2.2% | 1.5%(2) |
| Common stocks | 11.5% | 8.3% | 7.7%(3) |

3.8% - 3.8% = 0%(1)
5.3% - 3.8% = 1.5%(2)
11.5% - 3.8% = 7.7%(3)

$$Fisher\ equation: (1 + r\ nominal) = (1 + r\ real) \cdot (1 + i)$$

$$r\ real = \frac{(1 + \text{r nominal})}{(1 + \text{i})} - 1$$

# 4. Corporate Valuation (BMAE, Chapters 2, 4, 5, and 6)

## 4.1. True / False Questions

a. The only possible payoff to holders of common stocks are cash dividends.

   a. True
   **b. False**

b. The constant growth formula for stock valuation does not work for firms with negative dividend growth rates.

   a. True
   **b. False**

c. A large percentage of the total value of a growth stock comes from the present value of its growth opportunities.

   **a. True**
   b. False

d. A stock's price is based on the expected present value, at the appropriate discount rate, of all the stock's future earnings.

   a. True
   **b. False**

e. Due to the value additivity principle, the theoretical value of a share equals the present value of the dividends plus the present value of the expected sales revenue at the end of the holding period.

   **a. True**
   b. False

f. When the management decides to increase the payout ratio, investors can expect a higher growth rate of the firm.

   a. True
   **b. False**

g. If retained earnings can be invested at the same rate of return as the one obtained on the capital market and there are no frictions (as in a Modigliani-Miller world), the enterprise value is independent of the payout ratio.

   **a. True**
   b. False

h. According to the Gordon Growth Model, the expected return equals the dividend yield minus the expected rate of growth in dividends.

   a. True
   **b. False**

i. To calculate the market value of equity from the enterprise value, the market value of debt has to be subtracted from the enterprise value.

   **a. True**
   b. False

### 4.2. Market capitalization

Assume General Electric has about 10 billion shares outstanding and the stock price is USD 37. Also, assume the P/E ratio is about 18. What is the most likely market capitalization for GE?

a. USD 10 billion
b. USD 18.3 billion
c. USD 37 billion
**d. USD 370 billion**
e. USD 666 billion

$$MarketCap = 10 \cdot 37 = 370$$

### 4.3. Payout ratio

A newspaper quotes the following figures for a company: Dividend = USD 1.12, P/E ratio = 18.3, and price = USD 37.22. What is the most likely dividend payout ratio for the company?

a. 18%
b. 35%
c. 37%
d. 45%
**e. 55%**

$$Div\ Payout\ ratio = \frac{Dividend}{Earnings} = \frac{1.12}{\frac{37.22}{18.3}} = 0.55$$

### 4.4. Expected return

Super Computer Company's stock is selling for USD 100 per share today. It is expected that it will pay a dividend of USD 6 per share in exactly one year and then can immediately be sold for USD 114 per share. What is the most likely expected rate of return for shareholders selling the stock in exactly one year?

**a. 20%**
b. 15%
c. 10%
d. 25%
e. None of the answers a. to d. are correct.

$$E(r) = \frac{114 + 6 - 100}{100} = 0.2$$

### 4.5. Stock price

Deluxe Company expects to pay a dividend of USD 2 per share at the end of year 1, USD 3 per share at the end of year 2, and then be sold for USD 32 per share at the end of year 2. The required rate of return on the stock is 15%. What is the most likely current value of the stock?

a. **USD 28.20**
b. USD 32.17
c. USD 32.00
d. USD 29.18
e. USD 30.10

$$P = \frac{2}{1+0.15} + \frac{3+32}{(1+0.15)^2} = 28.20$$

## 4.6. Stock price

Casino Inc. expects to pay a dividend of USD 3 per share in exactly one year and the dividends are expected to grow at a constant rate of 6% per year forever. The required rate of return on the stock is 18%. What is the most likely current value of the stock today?

a. USD 16.67
b. USD 17.67
c. **USD 25.00**
d. USD 26.50
e. USD 50.00

$$PV = \frac{3}{0.18-0.06} = 25$$

## 4.7. Gordon Growth Model

The constant dividend growth formula $P_0 = Div_1/(r - g)$ assumes:
I) that dividends grow at a constant rate $g$, forever,
II) $r > g$;
III) $g$ is never negative

a. I only
b. II only
c. **I and II only**
d. III only
e. None of the answers a. to d. are correct.

## 4.8. Expected return

Will Co. is expected to pay a dividend of USD 2 per share in exactly one year, and the dividends are expected to grow at a constant rate of 4% p.a. forever. The current price of the stock is

USD 20 per share. What is the most likely expected return or the cost of equity capital for the firm?

a. 4%
b. 6%
c. 10%
**d. 14%**
e. 20%

$$r = \frac{2}{20} + 0.04 = 0.14$$

## 4.9. Expected cost of equity

One can estimate the expected rate of return or the cost of equity capital as follows:

a. Dividend yield - expected rate of growth in dividends
**b. Dividend yield + expected rate of growth in dividends**
c. Dividend yield/expected rate of growth in dividends
d. (Dividend yield) × (expected rate of growth in dividends
e. None of the answers a. to d. are correct.

## 4.10. Dividend growth rate

MJ Co. consistently pays out 60% of its earnings as dividends and can be expected to continue to do so. Its return on equity is 15% and can also be expected to stay at that level. What is the most likely stable dividend growth rate for the firm?

a. 2%
b. 5%
c. **6%**
d. 9%
e. 15%

$$g = 0.15 \cdot (1 - 0.6) = 0.06$$

### 4.11. Equity valuation

The In-Tech Co. just paid a dividend of USD 1 per share. Analysts expect its dividend to grow at 25% per year for the next three years and then to drop to 5% per year for infinity. The required rate of return on the stock is 18%. What is the most likely current value of the stock?

a. USD 11.93
b. USD 12.15
**c. USD 12.97**
d. USD 15.20
e. USD 15.78

$$P = \frac{1 \cdot (1+0.25)^1}{(1+0.18)^1} + \frac{1 \cdot (1+0.25)^2}{(1+0.18)^2} + \frac{1 \cdot (1+0.25)^3}{(1+0.18)^3} + \frac{1}{(1+0.18)^3} \cdot \frac{(1+0.25)^3 \cdot (1+0.05)}{0.18-0.05}$$

P = 12.97

### 4.12. Dividend growth rate

Ocean Co. is fully equity financed and just paid a dividend of USD 2 per share out of earnings of USD 4 per share. The book value per share is USD 25. What is the most likely expected growth rate of dividends?

a. 4%
**b. 8%**
c. 10%
d. 12%
e. 16%

$$g = ROE \cdot plowback\ ratio$$

$$g = \frac{4}{25} \cdot \left(1 - \frac{2}{4}\right) = 0.08$$

### 4.13. Expected stock return

River Co. is fully equity financed and just paid a dividend of USD 2 per share out of earnings of USD 4 per share. Its book value per share is USD 25 and its stock is currently selling for USD 40 per share. What is the most likely required rate of return on the stock?

a. 7.2%
b. 10.4%
**c. 13.4%**
d. 14.7%
e. 15.2%

$$P = \frac{D_1}{r-g} = \frac{D_0 \cdot (1+g)}{r-g} = \frac{D_0 \cdot (1 + (ROE \ \cdot plowback\, ratio))}{r - (ROE \ \cdot plowback\, ratio)}$$

$$r = \frac{2 \cdot \left(1 + \left(\frac{4}{25} \cdot \left(1 - \frac{2}{4}\right)\right)\right)}{40} + \left(\frac{4}{25} \cdot \left(1 - \frac{2}{4}\right)\right) = 0.134$$

## 4.14. Gordon Growth Model

Today, Z-Corporation has earnings per share (EPS) of EUR 10, whereof 70% are distributed to the shareholders. Over the past years, earnings and dividends increased by 5.5% per year and will grow at the same rate in the future. The cost of equity of Z-Corporation is 8.4%. What is the most likely current equity value (per share) of Z-Corporation under the Gordon Growth Model?

a. EUR 250.00
b. EUR 253.12
**c. EUR 254.66**
d. EUR 260.00
e. EUR 262.25

$$Equity\ per\ share = \frac{10 \cdot 0.7 \cdot (1 + 0.055)}{0.084 - 0.055} = 254.66$$

## 4.15. Growth stock

A high proportion of the value of a growth stock typically comes from:

a. past dividend payments
b. past earnings
**c. PVGO**
d. both a. and b.
e. None of the answers a. to d. are correct.

## 4.16. Present value of growth opportunities

Parcel Corporation expects to pay a dividend of USD 5 per share next year, and the dividend payout ratio is 50%. Dividends are expected to grow at a constant rate of 8% p.a. forever, and the required rate of return on the stock is 13% p.a. What is the most likely present value of the growth opportunities?

**a. USD 23.08**
b. USD 69.54
c. USD 76.92
d. USD 90.00
e. USD 100.00

$$EPS = \frac{5}{0.5} = 10 \rightarrow value\ with\ \boldsymbol{no}\ growth = \frac{10}{0.13} = 76.92$$

$$value\ with\ growth = \frac{5}{0.13 - 0.08} = 100$$

$$PVGO = 100 - 76.92 = 23.08$$

## 4.17. Horizon values in DCF valuation

A company forecasts growth in free cash flows of 6% for the next five years and 3% thereafter. Given last year's *FCF* was USD 100, what is its most likely horizon value, assuming that the company cost of capital is 8%?

a. USD 0
b. USD 1,672.46
c. USD 2,000.00
d. USD 2,676.39
**e. USD 2,756.74**

$$HV_5 = \frac{FCF_6}{r - g} = \frac{100 \cdot (1 + 0.06)^5 \cdot (1 + 0.03)}{0.08 - 0.03} = 2{,}756.74$$

## 4.18. Change in the value of a firm

The accumulated NPV of the free cash flows (*FCF*) of the first five years is 100. There will be a *FCF* of 30 in year 6. The cost of capital amounts to 6% and the constant growth rate in free cash flows is 3%. What is the most likely change in value of the firm, if due to deteriorating economic conditions the growth rate as of year 6 must be lowered by 100 basis points?

The value of the firm decreases by …

a. 20.48%
b. 21.89%
**c. 22.05%**
d. 22.73%
e. 25.00%

$$PV_{g=0.03} = \sum_{t=1}^{5} PV(FCF_t) + PV(HV) = 100 + \frac{1}{(1 + 0.06)^5} \cdot \frac{30}{0.06 - 0.03} = 847.26$$

$$PV_{g=0.02} = \sum_{t=1}^{5} PV(FCF_t) + PV(HV) = 100 + \frac{1}{(1 + 0.06)^5} \cdot \frac{30}{0.06 - 0.02} = 660.44$$

$$847.26 \cdot (1 - x) = 660.44 \rightarrow x = -0.2205$$

## 4.19. Multiples

The Wigros Corporation is thinking about going public. Estimate the equity value of Wigros, assuming that Wigros had sales of CHF 20 billion last year and that the average equity/sales multiple of the peer group was 0.6 last year.

The most likely equity value of Wigros is …

a. CHF 10.33 billion
b. CHF 11.00 billion
**c. CHF 12.00 billion**
d. CHF 23.33 billion
e. CHF 33.33 billion

$$\frac{Equity}{Sales} = 0.6 \rightarrow \frac{Equity}{20\ bn} = 0.6 \ \rightarrow Equity = 12\ bn$$

## 4.20. Growth-adjusted P/E ratio (PEG)

You find the following information about several firms in the same industry:

| Company Name | Trailing PE | Expected Growth | Standard Deviation |
|---|---|---|---|
| Alpha Inc. | 8.40 | 2.40% | 24.3% |
| Beta Ltd. | 20.16 | 16.80% | 28.9% |
| Gamma SA | 15.50 | 12.50% | 30.2% |
| Delta Corp. | 11.33 | 10.30% | 22.6% |
| Epsilon LLC | 15.48 | 12.90% | 26.1% |

Which statement is correct?

a. Alpha Inc. is definitively undervalued.
b. The high standard deviations of the expected growth rates make it impossible to calculate the growth-adjusted P/E ratio (PEG).
c. Considering the low growth-rate of Alpha Inc., Alpha Inc.'s growth-adjusted P/E ratio (PEG) ranks at the third place compared to the other industry peers.
**d. Considering the low growth-adjusted P/E ratio of Alpha Inc., Alpha Inc. looks overvalued compared to the Trailing P/E ratio of 8.40.**
e. None of the answers a. to d. are correct.

| Company Name | Trailing PE | Expected Growth | PEG | Implied P/E ratio (adj.) |
|---|---|---|---|---|
| Alpha Inc. | 8.4 | 2.40% | 3.5 | 3.9552 |
| Beta Ltd. | 20.16 | 16.80% | 1.2 | 27.6864 |
| Gamma SA | 15.5 | 12.50% | 1.24 | 20.6 |
| Delta Corp. | 11.33 | 10.30% | 1.1 | 16.9744 |
| Epsilon LLC | 15.48 | 12.90% | 1.2 | 21.2592 |
| Average | 14.174 | | 1.648 | |

$$PEG = \frac{Trailing\ P/E}{expected\ growth \cdot 100}$$

$$Implied\ PE\ ratio\ (adj.) = average\ PEG \cdot expected\ growth \cdot 100$$

### 4.21. Two-stage DCF model

Consider the following three stocks. Stock A is expected to provide a dividend of USD 10 a share forever. Stock B is expected to pay a dividend of USD 5 next year. Thereafter, dividend growth is expected to be 4% a year forever. Stock C is expected to pay a dividend of USD 5 next year. Thereafter, dividend growth is expected to be 20% a year for 5 years (i.e., years 2 through 6) and zero thereafter.

If the appropriate discount rate for each stock is 10%, which stock is the most valuable?

a. Stock A
b. Stock B
**c. Stock C**
d. Stock A, B, and C are equally valuable
e. None of the answers a. to d. are correct

$$PV_A = \frac{10}{0.10} = 100$$
$$PV_B = \frac{5}{0.10 - 0.04} = 83.33$$

$$PV_C = \frac{5}{1.1} + \frac{6}{(1.1)^2} + \frac{7.2}{(1.1)^3} + \frac{8.64}{(1.1)^4} + \frac{10.37}{(1.1)^5} + \frac{12.44}{(1.1)^6} + \frac{\left(\frac{12.44}{0.1}\right)}{(1.1)^6} = 104.47$$